

Coordinate Geometry — Solutions

Complete Step-by-Step Solutions with Trap Alerts & Key Insights

Question 1 [Easy]

Correct Answer: B) 2

Step-by-Step Solution:

- Slope = $(y_2 - y_1) / (x_2 - x_1) = (13 - 5) / (6 - 2)$
- $= 8 / 4 = 2$

Why each wrong option fails:

- A) $1/2$ — This is the reciprocal of the slope; likely computed run/rise instead of rise/run.
- C) 4 — This is just the run $(6 - 2)$; ignores the formula.
- D) 8 — This is the rise only $(13 - 5)$; forgot to divide by the run.

■ **TRAP ALERT:** Students who compute run/rise instead of rise/run get the reciprocal $1/2$.

■ **KEY INSIGHT:** Slope = rise / run. Always subtract in the same order for both coordinates.

■ **FAST METHOD:** Mental math: rise = 8, run = 4, slope = 2. Done in 5 seconds.

■ **VERDICT:** Hand-solve — pure arithmetic, faster than opening Desmos.

Question 2 [Easy]

Correct Answer: A) $y = 3x + 4$

Step-by-Step Solution:

- Use point-slope: $y - y_1 = m(x - x_1)$
- $y - 7 = 3(x - 1)$
- $y - 7 = 3x - 3$
- $y = 3x + 4$

Why each wrong option fails:

- B) $y = 3x + 7$ — Used $y = mx + y_1$ directly, ignoring the point-slope formula.
- C) $y = 3x + 10$ — Added 7 and 3 instead of subtracting properly.
- D) $y = 3x - 4$ — Sign error when distributing the 3.

■ **TRAP ALERT:** Plugging in $y_1 = 7$ directly as the y-intercept gives the wrong option B.

■ **KEY INSIGHT:** Always use $y - y_1 = m(x - x_1)$ and then simplify to slope-intercept form.

■ **FAST METHOD:** $y = 3(1) + b \rightarrow 7 = 3 + b \rightarrow b = 4 \rightarrow y = 3x + 4$.

■ **VERDICT:** Hand-solve — quick substitution; Desmos unnecessary.

Question 3 [Easy]

Correct Answer: A) (0, -5)

Step-by-Step Solution:

- Rewrite in slope-intercept form: $4x - 2y = 10$
- $-2y = -4x + 10$
- $y = 2x - 5$
- The y-intercept is (0, -5).

Why each wrong option fails:

- B) (0, 5) — Sign error when dividing by -2.
- C) (0, 10) — Read the constant from the original equation without solving.
- D) (0, -10) — Divided only the 10 by -2, not the 4x term.

■■ TRAP ALERT: Not dividing every term by -2 leads to sign errors.

■ KEY INSIGHT: Set $x = 0$ directly: $-2y = 10 \rightarrow y = -5$. No need to fully convert.

■ FAST METHOD: Plug $x = 0$: $4(0) - 2y = 10 \rightarrow y = -5$. Instant.

■🏠 VERDICT: Hand-solve — one substitution; 10 seconds max.

Question 4 [Easy]

Correct Answer: B) -3/4

Step-by-Step Solution:

- Slope = $(y_2 - y_1)/(x_2 - x_1) = (-6 - 0)/(8 - 0)$
- $= -6/8 = -3/4$

Why each wrong option fails:

- A) -4/3 — Computed rise/run as x/y instead of y/x (inverted fraction).
- C) 3/4 — Dropped the negative sign.
- D) 4/3 — Inverted AND dropped the negative.

■■ TRAP ALERT: Confusing which coordinate goes on top (run/rise vs rise/run) is the #1 slope error.

■ KEY INSIGHT: Through the origin simplifies things: slope = $y/x = -6/8 = -3/4$.

■ FAST METHOD: Since one point is (0,0), slope is simply -6/8 reduced to -3/4.

■🏠 VERDICT: Hand-solve — trivial with origin; no calculator needed.

Question 5 [Medium]

Correct Answer: -17

Step-by-Step Solution:

- $y - (-2) = 5(x - 3)$
- $y + 2 = 5x - 15$
- $y = 5x - 17$
- So $b = -17$.

■ ■ **TRAP ALERT:** Students who compute $-2 - 5(3) = -2 - 15 = -17$ correctly, but some get -13 by using +2 instead of $-(-2)$.

■ **KEY INSIGHT:** In $y = mx + b$, plug in the point: $-2 = 5(3) + b \rightarrow b = -2 - 15 = -17$.

■ **FAST METHOD:** $b = y - m \cdot x = -2 - 5(3) = -2 - 15 = -17$.

■ **VERDICT:** Hand-solve — simple arithmetic.

Question 6 [Easy]

Correct Answer: A) (2, 2)

Step-by-Step Solution:

- Test each point in $3x + 6y = 18$:
- A: $3(2) + 6(2) = 6 + 12 = 18$ ✓
- B: $3(3) + 6(2) = 9 + 12 = 21$ ✗
- C: $3(4) + 6(2) = 12 + 12 = 24$ ✗
- D: $3(6) + 6(2) = 18 + 12 = 30$ ✗

Why each wrong option fails:

- B) (3, 2) — Gives 21, not 18.
- C) (4, 2) — Gives 24, not 18.
- D) (6, 2) — Gives 30, not 18.

■ ■ **TRAP ALERT:** Students may confuse x and y coefficients or miscalculate products mentally.

■ **KEY INSIGHT:** Plug-and-check is the fastest approach when all options share the same y -value.

■ **FAST METHOD:** Since $y = 2$ in all options: $3x + 12 = 18 \rightarrow 3x = 6 \rightarrow x = 2 \rightarrow (2, 2)$.

■ **VERDICT:** Hand-solve — one equation, one unknown once you notice $y = 2$ throughout.

Question 7 [Medium]

Correct Answer: A) (7, -1)

Step-by-Step Solution:

- Midpoint formula: $M = ((x_1 + x_2)/2, (y_1 + y_2)/2)$
- $(1 + x_2)/2 = 4 \rightarrow x_2 = 7$
- $(7 + y_2)/2 = 3 \rightarrow y_2 = -1$
- Other endpoint = (7, -1)

Why each wrong option fails:

- B) (5, 5) — Averaged instead of doubling.
- C) (2.5, 5) — Found midpoint of midpoint and endpoint.
- D) (3, -4) — Subtracted coordinates in the wrong direction.

■■ TRAP ALERT: Some students find the midpoint of the midpoint and the given endpoint instead of solving for the other endpoint.

■ KEY INSIGHT: If $M = (mx, my)$, then the other endpoint = $(2mx - x_1, 2my - y_1)$.

■ FAST METHOD: Double the midpoint and subtract the known endpoint: $(8-1, 6-7) = (7, -1)$.

■ VERDICT: Hand-solve — direct formula application.

Question 8 [Medium]

Correct Answer: B) $x = -2$

Step-by-Step Solution:

- Undefined slope means a vertical line.
- Vertical lines have the form $x = \text{constant}$.
- Since the line passes through $(-2, 4)$, the equation is $x = -2$.

Why each wrong option fails:

- A) $y = 4$ — Horizontal line (slope = 0, not undefined).
- C) $y = -2x + 4$ — This has slope -2, which is defined.
- D) $x = 4$ — Wrong constant; used y-coordinate instead of x-coordinate.

■■ TRAP ALERT: Confusing 'undefined slope' (vertical) with 'zero slope' (horizontal) is extremely common.

■ KEY INSIGHT: Undefined slope = vertical line = $x = \text{constant}$. Zero slope = horizontal line = $y = \text{constant}$.

■ FAST METHOD: Vertical $\rightarrow x = (\text{x-coordinate of the point}) = -2$. Done.

■ VERDICT: Hand-solve — conceptual, no calculation needed.

Question 9 [Medium]

Correct Answer: A) $(\frac{9}{2}, 0)$

Step-by-Step Solution:

- At the x-axis, $y = 0$.
- $0 = -2x + 9$
- $2x = 9$
- $x = \frac{9}{2}$
- The x-intercept is $(\frac{9}{2}, 0)$.

Why each wrong option fails:

- B) $(9, 0)$ — Set $y = 0$ but forgot to divide by 2.
- C) $(-\frac{9}{2}, 0)$ — Sign error when solving.
- D) $(0, 9)$ — This is the y-intercept, not x-intercept.

■ ■ **TRAP ALERT:** Giving the y-intercept $(0, 9)$ when the question asks for the x-intercept is a classic trap.

■ **KEY INSIGHT:** x-intercept means $y = 0$. y-intercept means $x = 0$. Don't mix them up.

■ **FAST METHOD:** Set $y = 0$: $x = \frac{9}{2}$. Instant.

■ 🏠 **VERDICT:** Hand-solve — one-step algebra.

Question 10 [Medium]

Correct Answer: 10

Step-by-Step Solution:

- Distance = $\sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$
- = $\sqrt{[(7-1)^2 + (11-3)^2]}$
- = $\sqrt{36 + 64}$
- = $\sqrt{100} = 10$

■ ■ **TRAP ALERT:** Some students forget to square the differences or take the square root at the end.

■ **KEY INSIGHT:** Recognize 6-8-10 as a multiple of the 3-4-5 Pythagorean triple.

■ **FAST METHOD:** $6^2 + 8^2 = 36 + 64 = 100$. $\sqrt{100} = 10$. Recognize the Pythagorean triple.

■ 🏠 **VERDICT:** Hand-solve — Pythagorean triple recognition is faster than Desmos.

Question 11 [Medium]

Correct Answer: C) 12

Step-by-Step Solution:

- The line crosses the y-axis at (0, 4) and the x-axis at (6, 0).
- These intercepts, together with the origin, form a right triangle.
- Base = 6 (along the x-axis from the origin to (6, 0)).
- Height = 4 (along the y-axis from the origin to (0, 4)).
- Area = $(1/2)(6)(4) = 12$.

Why each wrong option fails:

- A) 6 — Forgot the $1/2$ in the area formula and only multiplied one pair.
- B) 10 — Added the intercepts instead of multiplying.
- D) 24 — Forgot to multiply by $1/2$.

■ ■ TRAP ALERT: Forgetting the $1/2$ factor in the triangle area formula gives 24 (option D).

■ KEY INSIGHT: For a triangle with vertices at (0,0), (a,0), (0,b): Area = $(1/2)|ab|$.

■ FAST METHOD: Area = $(1/2)(6)(4) = 12$. Memorize the intercept-triangle formula.

■ 🏠 VERDICT: Hand-solve — direct formula.

Question 12 [Medium]

Correct Answer: A) Slope = $5/3$, y-intercept = (0, -5)

Step-by-Step Solution:

- $5x - 3y = 15$
- $-3y = -5x + 15$
- $y = (5/3)x - 5$
- Slope = $5/3$, y-intercept = (0, -5).

Why each wrong option fails:

- B) Slope = $-5/3$ — Sign error when dividing $-5x$ by -3 .
- C) y-int = (0, 5) — Divided 15 by -3 incorrectly (missed the negative).
- D) Slope = $3/5$ — Inverted the fraction.

■ ■ TRAP ALERT: Dividing by a negative flips signs on ALL terms. Missing one sign is the #1 error.

■ KEY INSIGHT: Standard form $Ax + By = C$ converts to slope = $-A/B$, y-int = C/B .

■ FAST METHOD: Slope = $-A/B = -(5)/(-3) = 5/3$. y-int = $C/B = 15/(-3) = -5$.

■ 🏠 VERDICT: Hand-solve — use the $-A/B$ shortcut instead of full rearrangement.

Question 13 [Medium]

Correct Answer: 1

Step-by-Step Solution:

- Midpoint x-coordinate: $(a + 5)/2 = 3$
- $a + 5 = 6$
- $a = 1$
- Verify y: $(1 + 9)/2 = 5$ ✓

■ TRAP ALERT: Some students solve $(a + 5)/2 = 3$ as $a + 5 = 3$, forgetting to multiply by 2 first.

■ KEY INSIGHT: Midpoint formula in reverse: multiply midpoint coordinate by 2, then subtract the known endpoint.

■ FAST METHOD: $a = 2(3) - 5 = 1$.

■ VERDICT: Hand-solve — one-step equation.

Question 14 [Hard]

Correct Answer: C) 6

Step-by-Step Solution:

- Rewrite: $2x - ky = 14 \rightarrow y = (2/k)x - 14/k$
- Slope = $2/k = 1/3$
- Cross-multiply: $k = 6$

Why each wrong option fails:

- A) 2 — Set $2/k = 1$ instead of $1/3$.
- B) 3 — Confused k with the denominator of the slope.
- D) 7 — Used the constant 14 somehow.

■ TRAP ALERT: Students often forget the slope from standard form is $-A/B$, but here $-A/B = -2/(-k) = 2/k$.

■ KEY INSIGHT: From $Ax + By = C$, slope = $-A/B$. Here slope = $-2/(-k) = 2/k$.

■ FAST METHOD: $2/k = 1/3 \rightarrow k = 6$. Three seconds.

■ VERDICT: Hand-solve — simple proportion.

Question 15 [Hard]

Correct Answer: A) $y = -3/2 x + 7/2$

Step-by-Step Solution:

- Slope = $(-4 - 8)/(5 - (-3)) = -12/8 = -3/2$
- Using point $(-3, 8)$: $y - 8 = -3/2(x - (-3))$
- $y - 8 = -3/2(x + 3)$
- $y - 8 = -3/2 x - 9/2$
- $y = -3/2 x - 9/2 + 16/2$
- $y = -3/2 x + 7/2$

Why each wrong option fails:

- B) $y = -3/2 x + 11/2$ — Arithmetic error when combining fractions.
- C) $y = 3/2 x + 7/2$ — Wrong sign on slope.
- D) $y = 3/2 x + 11/2$ — Both slope sign and intercept wrong.

■ **TRAP ALERT:** Fraction arithmetic (converting 8 to 16/2) trips many students up.

■ **KEY INSIGHT:** Always convert whole numbers to fractions with the same denominator before adding.

■ **FAST METHOD:** Use Desmos to plot both points and read the equation. Faster when fractions are messy.

■ **VERDICT:** Desmos — plotting two points avoids fraction errors; read equation directly from graph.

Question 16 [Hard]

Correct Answer: C) 5

Step-by-Step Solution:

- Radius = distance from center to point on circle
- $r = \sqrt{[(7-3)^2 + (1-(-2))^2]}$
- $= \sqrt{16 + 9}$
- $= \sqrt{25} = 5$

Why each wrong option fails:

- A) 3 — Only calculated one leg of the distance.
- B) 4 — Only calculated the other leg.
- D) 7 — Added the legs instead of using Pythagorean theorem.

■ **TRAP ALERT:** Adding the differences ($4 + 3 = 7$) instead of using the distance formula.

■ **KEY INSIGHT:** Recognize 3-4-5 Pythagorean triple instantly. Distance = 5.

■ **FAST METHOD:** $\Delta x = 4$, $\Delta y = 3 \rightarrow$ 3-4-5 triple $\rightarrow r = 5$.

■ **VERDICT:** Hand-solve — Pythagorean triple recognition is instant.

Question 17 [Hard]

Correct Answer: 3

Step-by-Step Solution:

- Substitute (6, -2): $a(6) + 3(-2) = 12$
- $6a - 6 = 12$
- $6a = 18$
- $a = 3$

■ TRAP ALERT: Students sometimes plug coordinates into the wrong variables (x and y swapped).

■ KEY INSIGHT: Just substitute the point into the equation and solve for the unknown.

■ FAST METHOD: $6a - 6 = 12 \rightarrow 6a = 18 \rightarrow a = 3$.

■ VERDICT: Hand-solve — direct substitution.

Question 18 [Hard]

Correct Answer: A) 0

Step-by-Step Solution:

- If the two lines intersect at (2, 3), substitute into the second equation:
- $3(2) - 2(3) = c$
- $6 - 6 = c$
- $c = 0$

Why each wrong option fails:

- B) 3 — Only computed $3(2) - 3 = 3$.
- C) 6 — Only computed $3(2) = 6$.
- D) 12 — Added instead of subtracting: $6 + 6 = 12$.

■ TRAP ALERT: Students sometimes verify the point on the first line instead of substituting into the second.

■ KEY INSIGHT: If a point is on a line, substituting its coordinates must satisfy the equation.

■ FAST METHOD: $3(2) - 2(3) = 6 - 6 = 0$. One substitution.

■ VERDICT: Hand-solve — single substitution.

Question 19 [Hard]

Correct Answer: A) $10 + 2\sqrt{13}$

Step-by-Step Solution:

- $AB = \sqrt{[(5-1)^2 + (2-2)^2]} = \sqrt{16} = 4$ (horizontal leg)
- $BC = \sqrt{[(5-5)^2 + (8-2)^2]} = \sqrt{36} = 6$ (vertical leg)
- $CA = \sqrt{[(5-1)^2 + (8-2)^2]} = \sqrt{16 + 36} = \sqrt{52} = 2\sqrt{13}$
- Perimeter = $4 + 6 + 2\sqrt{13} = 10 + 2\sqrt{13}$

Why each wrong option fails:

- B) $10 + \sqrt{52}$ — Correct value but not simplified.
- C) $10 + 4\sqrt{13}$ — Incorrectly simplified $\sqrt{52}$.
- D) 20 — Estimated without calculating the hypotenuse.

■ TRAP ALERT: Not simplifying $\sqrt{52} = \sqrt{(4 \times 13)} = 2\sqrt{13}$.

■ KEY INSIGHT: Two sides are axis-aligned (easy lengths). Only the hypotenuse needs the distance formula.

■ FAST METHOD: Spot the right angle at B. Legs are 4 and 6. Hyp = $\sqrt{(16+36)} = 2\sqrt{13}$.

■ VERDICT: Hand-solve — right triangle with axis-aligned legs makes this manageable.

Question 20 [Hard]

Correct Answer: A) 6

Step-by-Step Solution:

- At the intersection: $2x - 3 = x^2 - 4x + c$
- $x^2 - 6x + (c + 3) = 0$
- For exactly one intersection, discriminant = 0:
- $(-6)^2 - 4(1)(c + 3) = 0$
- $36 - 4c - 12 = 0$
- $24 - 4c = 0$
- $c = 6$

Why each wrong option fails:

- B) 3 — Set up the equation incorrectly.
- C) 0 — Assumed c must be zero for tangency.
- D) -6 — Sign error in the discriminant calculation.

■ TRAP ALERT: Setting discriminant = 0 requires careful tracking of the rearranged quadratic's coefficients.

■ KEY INSIGHT: Exactly one intersection = discriminant $b^2 - 4ac = 0$ (tangency condition).

■ FAST METHOD: Set equations equal, rearrange, apply discriminant = 0 condition.

■ VERDICT: Hand-solve — discriminant formula is systematic; Desmos slider on c is an alternative.

■ TOPIC SUMMARY

- 1. SLOPE FORMULA:** $m = (y_2 - y_1)/(x_2 - x_1)$. Always rise/run. Common trap: computing run/rise.
- 2. POINT-SLOPE FORM:** $y - y_1 = m(x - x_1)$. Use when given a point and slope. Convert to slope-intercept for the answer.
- 3. STANDARD TO SLOPE-INTERCEPT:** $Ax + By = C \rightarrow \text{slope} = -A/B$, $y\text{-int} = C/B$. This shortcut saves time.
- 4. MIDPOINT FORMULA:** $M = ((x_1 + x_2)/2, (y_1 + y_2)/2)$. To find an endpoint: double the midpoint and subtract.
- 5. DISTANCE FORMULA:** $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. Know the Pythagorean triples: 3-4-5, 5-12-13, 8-15-17.
- 6. X-INTERCEPT vs Y-INTERCEPT:** $x\text{-int} \rightarrow \text{set } y = 0$. $y\text{-int} \rightarrow \text{set } x = 0$. Don't mix these up!
- 7. VERTICAL vs HORIZONTAL LINES:** Vertical: $x = c$ (undefined slope). Horizontal: $y = c$ (slope = 0).
- 8. TRIANGLE AREA WITH INTERCEPTS:** $\text{Area} = (1/2)|x\text{-int}||y\text{-int}|$ for axis triangles.
- 9. DISCRIMINANT FOR TANGENCY:** Line tangent to curve $\rightarrow$ set equal, rearrange to quadratic, set $b^2 - 4ac = 0$.
- 10. DESMOS vs HAND:** Use Desmos for ugly fractions, graphing intersections, and verifying. Hand-solve for Pythagorean triples, midpoints, and simple intercepts.